

**TSMC****OUTPERFORM**PRICE* TWD2,080 ■ TARGET PRICE **TWD2,520** (UPSIDE 21%)**TSMC ADR****OUTPERFORM**PRICE* USD382.7 ■ TARGET PRICE **USD478** (UPSIDE 25%)

FLASH NOTE

TSMC debuts A13 and updates tech roadmap

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Research Analyst & Publishing Entities

Alex Chang BNP Paribas Securities (Asia) Limited +852 28251849 alex.q.chang@asia.bnpparibas.com**What happened?**

TSMC delivered a robust technology roadmap update in 2026 North America Technology Symposium. Key announcements include the introduction of A13 process, A12 preview and expansion of CoWoS packaging capabilities.

A13 (HVM 2029): an optical shrink of A14, yielding 6% area savings while maintaining full backward compatibility. This reduces tape-out costs for clients, enabling faster migration.

A12 (HVM 2029): A12 is an enhancement from A14 platform. Previewed with “Super Power Retail” with backside power delivery for AI and HPC applications, A12 allows TSMC to segment its offering with premium performance.

N2U (HVM 2028): N2U further advances 2nm platform. Compared with N2P, N2U guarantees 3-4% speed gains or 8-10% power reduction compared to N2P, and 1.02-1.03x logic density improvement. It acts as a balanced, highly matured option for AI, HPC and mobile applications, leveraging established yield performance from 2nm platform.

Advanced packaging: TSMC is scaling its CoWoS to a 14-reticle size by 2028 (housing 10 large compute dies and 20 HBM stacks) to address the AI cluster bottlenecks.

TSMC – SoIC (HVM 2029): TSMC introduced A14-to-A14 SoIC stacking on its most advanced technology platform, delivering a 1.8x increase in die-to-die I/O density compared with N2-to-N2 SoIC and supporting higher bandwidth of data transfer between stacked chips.

Silicon photonics: TSMC – COUPE (co-packaged optics) promises 2x power efficiency and 10x latency reduction for data transfer between racks.

N2A: brings nanosheets to automotive for the first time and is scheduled to complete qualification in 2028, offering 15-20% speed gain over N3A. TSMC also makes “Auto-Use” design kit available within N2P to jumpstart development.

Specialty: N16HV introduces high voltage technology aimed for display drivers in various downstream applications. For smartphone display drivers, N16HV increases gate density by 41% and reduce power by 35% vs N28HV process. For near-eye displays, N16HV shrinks die area by 40% and reduce power by more than 20%

Meanwhile, TSMC has postponed adopting ASML’s latest High-NA EUV lithography machines until at least 2029 due to high price of USD400m and plans to maximize its existing EUV technology for upcoming advanced nodes.

BNPP View:

The comprehensive roadmap updates reaffirm TSMC’s undisputed technology leadership in the industry through the launch of leading-edge processes. Meanwhile, by strategically expanding its leading-edge portfolio to offer multiple nodes options, TSMC demonstrates a unique capability to cater to bespoke performance, power and cost needs of a highly diverse customer base.

Consequently, we firmly believe TSMC will remain the premier and most preferred foundry partner for leading global customers. As the structural AI build-out accelerates, we expect the company’s newly added advanced process capacity to experience sustained high demand, providing more revenue visibility and reinforcing its moat.

Figure 1: TSMC technology roadmap update

	A12 vs A14	A13 vs A14	A14 vs N2	N2U vs N2P	A16 vs N2P	N2P vs N2	N2 vs N3E	N3E vs N5	N3 vs N5	N5 vs N7
Speed Improvement @ Same Power	Higher for AI/HPC	Higher	+10%~15%	+3%~4%	+8% ~ 10%	+5%	10% ~ 15%	+18%	+10% ~ 15%	+15%
Power Reduction @ Same Speed	NA	Higher	+25%~30%	+8%~10%	-15% ~ 20%	-5% ~ 10%	-23% ~ -30%	-34%	-25% ~ -30%	-30%
Chip Density	NA	1.06x	~1.2x	1.02~1.03x	1.07~1.10x	NA	>1.1x	1.3x	NA	NA
HVM Start	2029	2029	2028	2028	2027	2H 2026	2H 2025	Q2/Q3 2023	2H 2022	2Q 2022

Source: TSMC, BNP Paribas Equity Research

Investment case, valuation and risks

TSMC (Outperform, Target Price TWD2,520)

Investment case

We have an Outperform rating on TSMC. We like TSMC for its superior tech strength, operating efficiency and highly visible orders from key customers, backing its full N3 process utilisation through 2026 and a smooth ramp-up of its N2 process. In addition, TSMC has established the most comprehensive ecosystem, expanding into advanced packaging, mask making, etc., which enables it to deliver solutions to almost any process or package in the most efficient and economical way. The more than doubled TAM under Foundry 2.0 indicates significant potential for TSMC's value market share expansion, paving the way for a c24% 2025-28E revenue CAGR. TSMC has planned to invest USD165b to set up fabs in the US. We think such moves will alleviate tariff risks, but will hurt TSMC's long-term margin & ROIC, as well as its technology edge over Intel. Despite the drag on TSMC's margins and ROIC in the long term from higher investment and operating costs in the US, we think the impact is lower than that from potential tariffs or investments in other US foundries.

Valuation methodology

We derive TSMC's target price of TWD2,520 based on 23x the average of our EPS estimates for 2026 and 2027. The valuation premium is justified by TSMC's technological edge and highly visible 27.5% 2025-28E earnings CAGR.

Risks

To the upside:

1) Stronger-than-expected demand from AI chips and high-end smartphone SoCs; 2) reduced geopolitical risks.

To the downside:

1) Weaker-than-expected demand from AI chips and high-end smartphone SoCs; 2) rising geopolitical risks.

TSMC ADR (Outperform, Target Price USD478)

Investment case

We have an Outperform rating on TSMC. We like TSMC for its superior tech strength, operating efficiency and highly visible orders from key customers, backing its full N3 process utilisation through 2026 and a smooth ramp-up of its N2 process. In addition, TSMC has established the most comprehensive ecosystem, expanding into advanced packaging, mask making, etc., which enables it to deliver solutions to almost any process or package in the most efficient and economical way. The more than doubled TAM under Foundry 2.0 indicates significant potential for TSMC's value market share expansion, paving the way for a c24% 2025-28E revenue CAGR. TSMC has planned to invest USD165b to set up fabs in the US. We think such moves will alleviate tariff risks, but will hurt TSMC's long-term margin & ROIC, as well as its technology edge over Intel. Despite the drag on TSMC's margins and ROIC in the long term from higher investment and operating costs in the US, we think the impact is lower than that from potential tariffs or investments in other US foundries.

Valuation methodology

We derive TSMC (ADR)'s target price of USD478 by assigning a 20% premium over TSMC's TP in TWD, in line with its past 1-year average premium.

Risks

To the upside:

1) Stronger-than-expected demand from AI chips and high-end smartphone SoCs; 2) reduced geopolitical risks.

To the downside:

1) Weaker-than-expected demand from AI chips and high-end smartphone SoCs; 2) rising geopolitical risks.

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TSMC ADR	TSM US	3; 4; 7; 8; 9; 10

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	Counts	%*	Counts	%
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Neutral (Hold)	428	34	101	24
Underperform (Sell)	202	16	42	21

* Please note that the percentages might not add up to 100% because of rounding.

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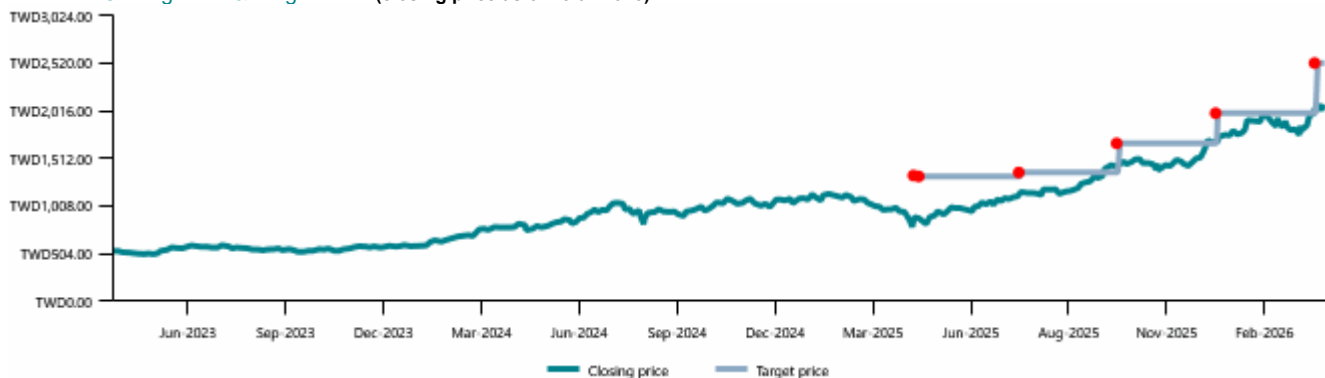
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Price and Ratings Chart

TSMC

Historical Closing Price & Target Price (closing price as of 23/04/2026)



Source: BNP Paribas

Historical rating & target price changes

Date & Time of Dissemination (London time)	Rating (1)	Target Price	Closing Price	Key Changes	Person(s) Involved
16 Apr. 2026 11:53	+	TWD2520.00	TWD2085.00	TP up	A. Chang, S. GE, G. Wang
15 Jan. 2026 11:37	+	TWD1990.00	TWD1690.00	TP up	A. Chang, S. GE, G. Wang
16 Oct. 2025 15:38	+	TWD1670.00	TWD1485.00	TP up	A. Chang, G. Wang
18 Jul. 2025 00:26	+	TWD1361.00	TWD1155.00	TP up	A. Chang
17 Apr. 2025 17:26	+	TWD1320.00	TWD847.00	TP down	A. Chang, C. Tang
10 Apr. 2025 10:05	+	TWD1330.00	TWD785.00	Initiation of coverage	A. Chang, C. Tang

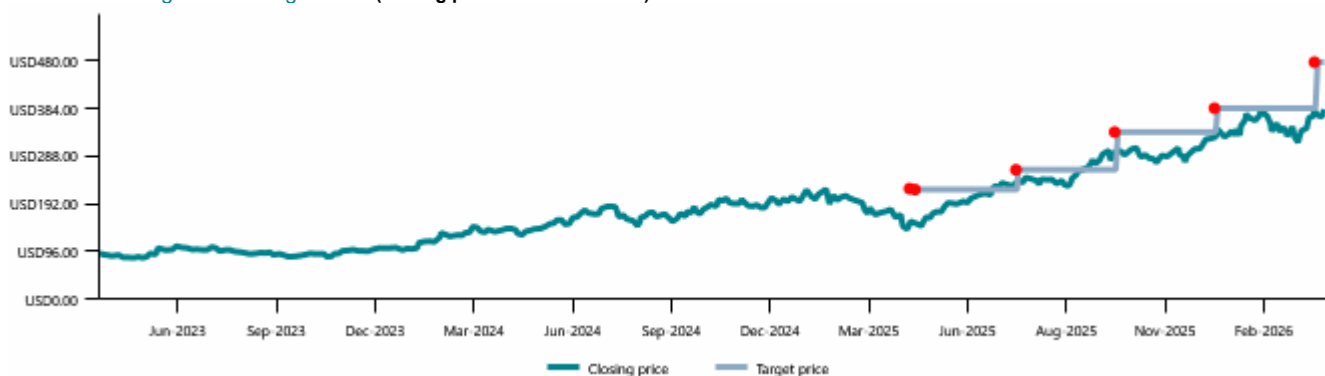
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TSMC ADR

Historical Closing Price & Target Price (closing price as of 23/04/2026)



Source: BNP Paribas

Historical rating & target price changes

Date & Time of Dissemination (London time)	Rating (1)	Target Price	Closing Price	Key Changes	Person(s) Involved
16 Apr. 2026 11:53	+	USD478.00	USD363.35	TP up	A. Chang, S. GE, G. Wang
15 Jan. 2026 11:37	+	USD385.00	USD341.64	TP up	A. Chang, S. GE, G. Wang
16 Oct. 2025 15:38	+	USD337.00	USD299.84	TP up	A. Chang, G. Wang
18 Jul. 2025 00:26	+	USD261.00	USD240.40	TP up	A. Chang
17 Apr. 2025 17:26	+	USD221.00	USD151.74	TP down	A. Chang, C. Tang
10 Apr. 2025 10:05	+	USD223.00	USD158.75	Initiation of coverage	A. Chang, C. Tang

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